Critical and Period-Two Controller Dynamics in Monomial RMSProp

Abstract

For the next-gradient variance ordering of RMSProp, we give exact critical-parameter asymptotics for the quartic trajectory initialized at $(x_0, v_0) = (2, 64)$, a constructive local theorem for every even monomial degree, and a certified sextic initialization. We also prove phase selection and positive parity-dependent physical amplitudes for a quartic controller two-cycle below the critical parameter. Finite-prefix certificates use outward-rounded rational arithmetic. Physical convergence already present in the source is distinguished from the sharper initialized asymptotic statements proved here.

Keywords: RMSProp; critical controller dynamics; monomial objectives; period-two phase selection; asymptotic amplitudes; rational interval certificates

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1 Algorithm and scope

For $f(x) = x^k/k$, even $k \geq 4$, consider the literal recurrence

$$x_{n+1} = x_n - \frac{x_n^{k-1}}{\sqrt{v_n}}, \qquad v_{n+1} = \beta v_n + (1-\beta)x_{n+1}^{2k-2}. \tag{1}$$

Thus variance uses the *next* gradient, $\varepsilon = 0$, and there is no bias correction. This is equation (28) of Bai et al. [1]. The initialization $v_0 = g_0^2$ is discussed in their Appendix E. Particular positive values of x_0 below are specified here, rather than attributed as uniquely prescribed by that paper.

Put $m = k - 2$, $w = x^2$, and $u = x^m/\sqrt{v}$. On $x, v > 0$, $0 < u < 1$, the exact state transformation is

$$w' = (1-u)^2 w, \qquad u' = \frac{g(u)}{\sqrt{1 + a(u)w}}, \tag{2}$$

$$g(u) = \beta^{-1/2} u(1-u)^m, \qquad a(u) = (\beta^{-1} - 1)u^2(1-u)^{2k-2}. \tag{3}$$

Here and below u' in (2) denotes the new controller. Indeed $x^{2k-2}/v = x^2u^2$. Conversely $x = \sqrt{w}$, $v = w^m/u^2$, so this is the full physical recurrence and not just a boundary surrogate.

2 The initialized quartic critical trajectory

Theorem 1. *In* (1)*, let* $k = 4$*,* $\beta = 1/16$*,* $x_0 = 2$*,* $v_0 = 64$*. Every iterate is positive and* $0 < u_n < 1$*. There exists* $C \in (0, \infty)$ *such that*

$$\sqrt{n}(-1)^n(u_n - 1/2) \longrightarrow 1/4, \qquad 2^n x_n \longrightarrow C, \qquad 16^n v_n \longrightarrow 4C^4.$$

In particular $x_{n+1}/x_n \to 1/2$. From index 12, the even controllers exceed $1/2$ and the odd controllers lie below it.

Proof. Here $g(u) = f(u) = 4u(1-u)^2$ and $a(u) = 15u^2(1-u)^6$. The rational interval prefix described in Appendix A, starting from $(u_0, w_0) = (1/2, 4)$, proves

$$\frac{5236}{10000} < u_{12} < \frac{5238}{10000}, \qquad 0 < w_{12} < \frac{3}{10^7} < \left(\frac{236}{10000}\right)^4.$$

Every prefix controller is in $(0,1)$. We now prove the entire infinite tail, including the continuous-region estimates needed by the certificate.

Set $h = 3/100$, $z = u - 1/2$, and $\mathcal{W} = \{0 < z \le h,\ 0 < w \le z^4\}$. Since $0 \le 1 - (1+r)^{-1/2} \le r/2$ for $r \ge 0$,

$$|u_1 - f(u)| \le (15/2)w.$$

On $u \in [1/2, 53/100]$, f decreases and $f(53/100) = 0.468308$. Consequently $46/100 < u_1 < 1/2$ and $w_1 \le w/4$. On the interval between u_1 and $f(u)$, $|f'| \le 2$, whence

$$|u_2 - f(f(u))| \le (135/8)w < 17w.$$

Exact composition gives

$$f(f(1/2+z)) - 1/2 = z - 16z^3 - 16z^4 + 32z^5 + 128z^6 - 384z^8 + 256z^9.$$

The coefficient of the cubic decrement is between 15 and 17 on $[0, h]$: this follows by bounding the remaining terms by their absolute coefficients at h. Since $17h = 51/100$, we obtain

$$z - 18z^3 \le z_2 \le z - 14z^3, \qquad 0 < w_2 < w/8. \tag{4}$$

Also $z_2 \ge z(1 - 18h^2) > (98/100)z > 0$, and $(98/100)^4 > 1/8$ implies $w_2 < z_2^4$. Thus $\mathcal{W}$ is two-step invariant and the prefix enters it.

Write $z_j = u_{12+2j} - 1/2$, $W_j = w_{12+2j}$. Then $z_j \downarrow 0$ and $W_j \le 8^{-j} W_0$. For $0 \le s \le h^2$,

$$(1 - 18s)^{-2} \le 1 + 40s,$$

because $(1-18s)^2(1+40s) - 1 = s(4 - 1116s + 12960s^2) \ge 0$. Therefore $z_j \ge (z_0^{-2} + 40j)^{-1/2}$ and $W_j / z_j^3 \to 0$. The exact composition and perturbation estimate now imply

$$z_{j+1} = z_j - 16z_j^3 + o(z_j^3), \qquad z_{j+1}^{-2} - z_j^{-2} \to 32.$$

Averaging increments yields $\sqrt{j}\, z_j \to 1/\sqrt{32}$. The one-step expansion $f(1/2+z) - 1/2 = -z - 2z^2 + 4z^3$ supplies the opposite odd phase and the full signed limit.

Finally $\sum_n w_n < \infty$. The exact variance identity gives

$$V = 64 \prod_{n=0}^{\infty} (1 + a(u_n) w_n) \in (0, \infty), \qquad 16^n v_n \to V,$$

since $0 \le a \le 15$ and $\log(1 + aw) \le 15w$. As $x_n^2 = u_n \sqrt{v_n}$, $(2^n x_n)^2 \to \sqrt{V}/2$. Thus $C = V^{1/4}/\sqrt{2}$ is finite and positive. Positivity and the phase statement follow throughout from the prefix and invariant region. ∎

3 A constructive local theorem for all even degrees

Let $s = 2/k$, $\rho = 1 - s$, and $\beta = \rho^{2m}$. Then $g(u) = \rho^{-m} u (1 - u)^m$ has $g(s) = s$, $g'(s) = -1$. Direct differentiation gives

$$g(s + z) - s = -z + a_2 z^2 + a_3 z^3 + O(z^4),$$
$$a_2 = -\frac{k}{k - 2}, \qquad a_3 = \frac{k^2 (k - 3)(k + 2)}{6(k - 2)^2},$$
$$g(g(s + z)) - s = z - c z^3 + O(z^4), \quad c = \frac{k^3 (k - 1)}{3(k - 2)^2} > 0.$$

The last identity follows by composition, whose cubic coefficient is $-2(a_2^2 + a_3)$.

Theorem 2. *For each even $k \ge 4$ one can construct a rational $h > 0$ such that any physical state satisfying $0 < u_N - s \le h$ and $0 < w_N \le (u_N - s)^4$ has a positive infinite orbit and*

$$\sqrt{n} (-1)^{n-N} (u_n - s) \to c^{-1/2}, \qquad \rho^{-n} x_n \to C \in (0, \infty), \qquad \beta^{-n} v_n \to C^{2m}/s^2.$$

This is a local entry theorem, not an all-initializations result.

Proof. The following finite construction specifies h. Put

$$d = \min(s, 1 - s)/4, \quad G = \rho^{-m}, \quad A = \beta^{-1} - 1,$$

$$B = GA/2, \quad q = (1 - s + d)^4 < 1, \quad D = 3c/2.$$

Expand the finite polynomials

$$g(s + z) - s = -z + \sum_{j \ge 2} t_j z^j, \quad g(g(s + z)) - s = z - c z^3 + \sum_{j \ge 4} p_j z^j,$$

and $g'(s + z) = \sum_j l_j z^j$. Define

$$L = \sum_j |l_j| d^j, \quad E = B(L + 1), \quad H(h) = \sum_{j \ge 2} |t_j| h^{j-1}, \quad R(h) = \sum_{j \ge 4} |p_j| h^{j-3}.$$

Starting at $h = d$, halve until

$$H(h) \le 1/2, \quad 2h + Bh^4 \le d, \quad R(h) + Eh \le c/2, \qquad (5)$$
$$Dh^2 \le 1/2, \qquad (1 - Dh^2)^4 > q.$$

All comparisons are rational, and termination follows as $h \downarrow 0$. For $0 \le u \le 1$, $0 \le g \le G$, $0 \le a \le A$, and $0 \le g(u) - u' \le Bw$. On $\mathcal{W} = \{0 < z \le h, 0 < w \le z^4\}$, the first test gives $-3z/2 \le g(s+z) - s \le -z/2$. The next test keeps the intermediate controller in $J = [s-d, s+d]$, below s. Since $|g'| \le L$ on J,

$$|u_2 - g(g(u))| \le LBw + Bw_1 \le Ew.$$

Equations (5) therefore yield

$$z - Dz^3 \le z_2 \le z - (c/2)z^3, \qquad w_2 \le qw.$$

Since $z_2 \ge z(1 - Dh^2) > 0$ and $(1 - Dh^2)^4 > q$, the region is two-step invariant. For its two-step iterates z_j, W_j, we have $z_j \downarrow 0$, $W_j \le q^j W_0$. The inequality $(1-t)^{-2} \le 1 + 8t$ on $[0, 1/2]$ gives $z_j \ge (z_0^{-2} + 8Dj)^{-1/2}$. Hence $W_j / z_j^3 \to 0$, $z_{j+1} = z_j - cz_j^3 + o(z_j^3)$, and $z_{j+1}^{-2} - z_j^{-2} \to 2c$. This proves the signed limit, including the opposite intermediate phase. As before $\sum w_n < \infty$, and

$$V = \beta^{-N} v_N \prod_{n \ge N} (1 + a(u_n) w_n) \in (0, \infty), \qquad \beta^{-n} v_n \to V.$$

Now $x_n^m = u_n \sqrt{v_n}$ and $\sqrt{\beta} = \rho^m$ give $\rho^{-n} x_n \to (s\sqrt{V})^{1/m}$. ∎

Corollary 3.1. *For $k = 6$, $\beta = 256/6561$, $x_0 = 3$, $v_0 = 3^{10}$,*

$$\sqrt{n}(-1)^n (u_n - 1/3) \to \sqrt{2/45}, \qquad (3/2)^n x_n \to C > 0, \qquad (6561/256)^n v_n \to 9C^8.$$

Proof. The choice $h = 1/100$ passes (5) with

$$d = 1/12, \quad c = 45/2, \quad B = 510705/8192, \quad L = 6005/4096,$$
$$E = 5158631205/33554432, \quad q = 81/256, \quad D = 135/4.$$

The exact prefix from $(u_0, w_0) = (1/3, 9)$ gives

$$97/10000 < u_{28} - 1/3 < 98/10000 < h, \quad 0 < w_{28} < 16/10^{10} < (97/10000)^4.$$

All prefix controllers belong to $(0, 1)$, and index 28 is even. Theorem 2 applies with the asserted phase. ∎

4 A quartic controller two-cycle below the threshold

Theorem 3. *Let $k = 4$, $\beta = 16/289$, $(x_0, v_0) = (2, 64)$, and*

$$a = \frac{1 - 1/\sqrt{17}}{2}, \qquad b = \frac{1 + 1/\sqrt{17}}{2}.$$

Every physical iterate is positive. For some $V > 64$,

$$u_{2j} \to b, \quad u_{2j+1} \to a, \quad \beta^{-n} v_n \to V,$$
$$\beta^{-2j/4} x_{2j} \to V^{1/4}\sqrt{b}, \qquad \beta^{-(2j+1)/4} x_{2j+1} \to V^{1/4}\sqrt{a}.$$

In particular the physical state converges to zero, whereas the controller has two distinct limiting phases. The one-step physical ratios tend to a at even indices and b at odd indices.

Proof. Here $r = 17/4$, $g(u) = ru(1-u)^2$, $A(u) = (273/16)u^2(1-u)^6$. On $(0,1)$, $0 < g(u) \le 17/27 < 1$, so (2) preserves positivity and $0 < u < 1$ globally. Also $0 \le g(u) - u' \le 9w$. Since $a + b = 1$, $ab = 4/17 = 1/r$,

$$g(a) = b, \quad g(b) = a, \quad g'(a)g'(b) = 9 - 2r = 1/2.$$

These boundary identities alone do not establish initialized attraction. The rational prefix proves

$$62065/100000 < u_{28} < 62066/100000, \quad 0 < w_{28} < 2/10^{17}.$$

The elementary bounds $24253/100000 < 1/\sqrt{17} < 24254/100000$, certified by squaring, imply $|u_{28} - b| < 10^{-3}$.

Set $h = 10^{-3}$, $W = 10^{-5}$, $\mathcal{R} = \{|u - b| \le h, 0 < w \le W\}$, $J_H = [.620, .623]$, $J_L = [.377, .381]$ (terminating decimals denote exact rationals). On these intervals $|g'| < 3/2$ and $|g'| < 1/2$, respectively; these follow directly from $|g'| = r(1-u)(3u-1)$ by endpoint product bounds. A state in $\mathcal{R}$ satisfies

$$a - (3/2)h - 9W \le u_1 \le a + (3/2)h,$$

whose endpoints lie strictly in J_L. Furthermore

$$w_1 \le (.38)^2 w < w/6, \qquad w_2 \le (.623)^2(.38)^2 w < w/16.$$

The two-step error is at most $(9/2)w + 9w_1 \le 6w$, whereas $|g(g(u)) - b| \le (3/4)|u - b|$. Therefore

$$|u_2 - b| \le (3/4)|u - b| + 6w \le (3/4)h + 6W < h.$$

This is a strict two-step trap. For $U_j = u_{28+2j}$, $W_j = w_{28+2j}$, iteration gives

$$|U_j - b| \le (3/4)^j \big(|U_0 - b| + (96/11)W_0\big), \qquad W_j \le 16^{-j} W_0.$$

Hence the even controllers tend to b and odd controllers to a. Since $\sum w_n < \infty$ and $A(u) < 18$,

$$V = 64 \prod_{n \geq 0} (1 + A(u_n) w_n) \in (64, \infty)$$

is the variance amplitude. The identity $(\beta^{-n/4} x_n)^2 = u_n \sqrt{\beta^{-n} v_n}$ proves the parity amplitudes. Finally $x_{n+1}/x_n = 1 - u_n$ proves the ratio statement. ∎

5 Relation to previous results and corrections

Bai et al. [1], Theorem 5.7, establish a strict-threshold ratio regime. Their Appendix B, Lemma B.1, already implies physical convergence in the quartic endpoint example, since $1/4 > 4/27$ and $u_0 < 1$. Physical convergence at equality is therefore not a new claim here. Their Appendix G also reports period doubling. The additional content is exact source-compatible initialization, phase selection, sharp controller constants, and finite positive physical amplitudes for the actual two-state recurrence.

A specific endpoint qualification is necessary. Proposition D.1 of [1], for the auxiliary scalar map, presents a strict derivative condition as an "if and only if" local asymptotic stability condition. Linearization does not establish necessity at modulus one; the explicit quartic identity above has two-step decrement $-16z^3 + O(z^4)$ and proves attraction at equality. Thus a blanket endpoint exclusion is false for this scalar map. This does not invalidate the source's separately strict-threshold theorem.

Ma, Wu and E [2] discuss quadratic physical cycles and optimizer oscillations; Bock and Weiß [3] construct quadratic cycles and study bifurcations. A physical nonzero cycle differs from the vanishing physical state and periodic controller here. Their results do not supply the initialized critical asymptotic constants proved above. These comparisons are bounded source-specific statements, not a literature-wide priority assertion. "Below threshold" here denotes a parameter inequality; it is not a claim of a technically subcritical flip bifurcation. No all-k global-initialization theorem, positive-ε extension, software-default initialization, or floating-point asymptotic theorem is asserted.

A Exact finite certificates

The accompanying scripts are `quartic_prefix.py`, `even_degree.py`, and `quartic_cycle.py`. These are standalone Python 3 certificates using only integers, `Fraction`, and `isqrt`. Each has a matching JSON trace. Run each script in the certificate directory. All rational trap inequalities and every prefix step are checked, not sampled.

For denominator $D = 10^{40}$, arithmetic is rounded down at lower endpoints and up at upper endpoints. Products take extrema of four endpoint products. For rational $q \geq 0$,

$$D^{-1} \operatorname{isqrt}(\lfloor q D^2 \rfloor) \leq \sqrt{q} < D^{-1} \big(\operatorname{isqrt}(\lfloor q D^2 \rfloor) + 1 \big).$$

Positive reciprocal endpoints are reversed. These elementary inclusion properties prove by induction that every recorded interval contains its true state. The continuous-region arguments in the proofs, rather than finite trajectory data alone, establish infinite-time conclusions. The general-degree script implements the halving construction, but its validity for all even degrees follows from Theorem 2, not finite test cases.

References

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